

## The Impact of Financial Education on Adolescents' Intertemporal Choices<sup>†</sup>

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*We study the impact of financial education on intertemporal choice in adolescence. The educational program was randomly assigned among high school students, and choices were measured using an incentivized experiment. Students who participated in the program make more time-consistent choices; are more likely to allocate payments to a single payment date, as opposed to spreading payment across two dates; and display increased consistency of choice with the law of demand. These findings suggest that financial education increases the quality of intertemporal decision-making and decreases narrow bracketing. (JEL C93, D14, D15, I21, J13)*

Many important financial decisions are intertemporal: saving for retirement, credit card debt, and mortgage choice, among others. As financial products have become more complex and the responsibility for retirement provision is increasingly being shifted towards individual households around the world, concerns about poor intertemporal decision-making intensify. A large body of evidence shows that many individuals save too little for retirement (e.g., Lusardi and Mitchell 2007, 2008), while others accumulate high amounts of credit card debt (e.g., Meier and Sprenger 2010). Such behaviors have been associated with a lack of financial knowledge (e.g., Hastings, Madrian, and Skimmyhorn 2013; Lusardi and Mitchell 2014; and Lusardi and Tufano 2015). More broadly, low wealth accumulation and wealth inequality have been attributed to poor financial knowledge among certain segments of the population (e.g., Lusardi, Michaud, and Mitchell 2017).

A central policy implication arising from these findings is that improvements in financial education may yield significant welfare increases. Indeed, this

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<sup>†</sup>Go to <https://doi.org/10.1257/pol.20170012> to visit the article page for additional materials and author disclosure statement(s) or to comment in the online discussion forum.

motivation has spurred a substantial increase in the provision of financial education programs in recent years.<sup>1</sup> Yet, the causal effect of financial education on financial decision-making is poorly understood.

Financial education may improve the quality of decision-making by increasing individuals' knowledge about the availability of saving and borrowing instruments. From this perspective, financial education programs might affect intertemporal choice by increasing sophistication in making these choices and thus help individuals implement their preferences. At the same time, financial education may affect deep time preferences. Becker and Mulligan (1997) suggest that educational programs that emphasize the future, such as financial education, may increase an individual's preference for the future. Especially at a young age, elevating an individual's focus on financial planning and achieving future savings goals could decrease impatience and present bias.

In this paper, we examine the effect of financial education on intertemporal choice among adolescents. Targeting financial education programs at adolescents and delivering them in schools holds particular appeal for two reasons: it allows universal population outreach, and it is timely, as experimental studies of time, risk, and social preferences show that preferences gradually stabilize at the end of adolescence.

We conduct a field experiment among over 900 students in 25 high schools, in which participation in a financial education program is randomly assigned. We then elicit incentivized intertemporal choices, using the Convex Time Budget (CTB) task (Andreoni and Sprenger 2012). The CTB allows researchers to assess whether and by how much subjects smooth time-dated monetary rewards<sup>2</sup> and whether their choices are consistent with the law of demand, a measure of the quality of decision-making (Giné et al. 2018).

We identify two main changes in intertemporal choice due to financial education. First, the treatment decreases time inconsistency in students' experimental choices. Treated students are less likely to allocate a larger share of the budget to the sooner payment date when it is immediate, a measure of present bias within the task. Second, the treatment leads to an increase of almost 20 percent in the frequency with which the entire budget is allocated to a single payment date. In addition, we observe an increase in the share of students who choose to allocate the entire budget to the earlier payment date at low interest rates and then switch to allocating the entire budget to the later payment date at higher interest rates.

The treatment leads to changes in two other dimensions of choice, which are related to the understanding of intertemporal choices. We observe a significant increase in the rate of consistency with the law of demand: treated students are more likely to decrease the allocation to the earlier payment date as the interest rate

<sup>1</sup>For example, in 2013, an estimated \$670 million was spent on such programs (CFPB 2013).

<sup>2</sup>The CTB task allows individuals to spread the payments offered by the experimenter across two payment dates or to allocate the entire payment either to the sooner or to the later payment date. Previous experimental tasks using choice lists only allow corner choices, i.e., individuals allocate the entire budget either to the sooner or to the later payment date. Early studies among adults include Collier and Williams (1999) and Harrison, Lau, and Williams (2002), while the first study among children was conducted by Bettinger and Slonim (2007). To allow for concave utility when estimating time-preference parameters, several studies have elicited risk preferences separately (e.g., Andersen et al. 2008, and Sutter et al. 2013).

increases, an indication of increased quality of decision-making (Giné et al. 2018). And, treated students are significantly less likely to exhibit negative discounting. This leads to an apparent increase in impatience (or delay sensitivity) among treated students. This increase can be explained by the fact that treated students are more likely to behave as assumed in standard models of intertemporal choice, where discount rates are zero or positive (see also Chakraborty et al. 2017).

The decrease in time inconsistency is robust to accounting for choice inconsistencies, suggesting that changes in understanding do not directly explain the effect of financial education on time consistency. At the same time, the observed decrease in time inconsistency paired with the increase in allocations of the budget to a single payment date is consistent with an increase in students' consideration of alternative sources of consumption when making intertemporal choices. That is, our findings suggest that treated students are less likely to treat experimental payments as consumption, ignoring rescheduling opportunities available through savings or borrowing vehicles (e.g., Frederick, Loewenstein, and O'Donoghue 2002; Harrison, Lau, and Williams 2002; Cubitt and Read 2007; Chabris, Laibson, and Schuldt 2008; and Sprenger 2015). When individuals are more "financially sophisticated" and integrate experimental payments with other sources of consumption, they are more likely to make time-consistent choices and less likely to smooth intertemporal payments within the task, two behaviors that treated students display.<sup>3</sup>

Taken together, our findings suggest that financial education changes intertemporal decision-making by increasing sophistication and understanding of intertemporal choice. This is a dimension of financial education which has received little attention so far (Ambuehl, Bernheim, and Lusardi 2015). Existing evaluations have mainly focused on changes in behaviors, such as savings, and often find no effects among adults (see Hastings, Madrian, and Skimmyhorn 2013; and Fernandes, Lynch, and Netemeyer 2014 for reviews),<sup>4</sup> while results are mixed for adolescents and young adults.<sup>5</sup> Consistent with these findings, survey measures of savings in our data do not indicate a change in savings behavior in response to the program.

Our conclusion is that financial education influences adolescents' intertemporal choices by making them more consistent with standard assumptions of discounted utility models (Samuelson 1937).<sup>6</sup> These positive effects of financial education on the quality of intertemporal decision-making could have important long-run consequences for an individual's financial situation. For example, there is evidence

<sup>3</sup>Further, we examine the correlation between external savings behavior and experimental choices among adolescents. In several studies, such a correlation has been considered an indication of the validity of experimental tasks in identifying individual time preferences (e.g., Castillo et al. 2011; and Sutter et al. 2013). This relationship is weakened among treated students, suggesting that the program makes experimental choices less informative about time-preference parameters.

<sup>4</sup>Recent studies suggest that the characteristics of the program may be crucially important. Combining education with grants improves the outcomes of male entrepreneurs (Berge, Bjorvatn, and Tungodden 2015), as does teaching simple financial heuristics (Drexler, Fischer, and Schoar 2014) or using different visual tools (Lusardi et al. 2017).

<sup>5</sup>An increasing number of programs and researchers focuses on financial education among youth, for which there are potentially large long-term effects and which take place before individuals have made important irreversible financial decisions (Lusardi, Mitchell, and Curto 2010). The evidence on the impact of such programs, however, is mixed (Bernheim, Garrett, and Maki 2001; Bruhn et al. 2013; Becchetti, Caiazza, and Coviello 2013; Cole, Paulson, and Shastry 2013; Berry, Karlan, and Pradhan 2015; Lührmann, Serra-Garcia, and Winter 2015).

<sup>6</sup>For an approach based on testing the understanding of opportunity sets, see Ambuehl, Bernheim, and Lusardi (2015).

that better decision-making is associated with higher levels of wealth accumulation (Choi et al. 2014).

Our findings also contribute to the debate on the identification of individual time preferences in experiments using time-dated monetary payments. Students in the control group display significant time inconsistency. The estimated present-bias parameter for this group is close to that found when intertemporal choices are elicited in the effort domain (Augenblick, Niederle, and Sprenger 2015). This suggests that using monetary rewards to elicit time preferences can be a valid approach under specific conditions. To the best of our knowledge, our study is the first to identify an important factor, financial education, that may limit the task's validity.<sup>7</sup>

The remainder of the paper proceeds as follows. In the next section, we describe the financial education program. In Section II, we describe the experimental task, the methods used, and the conceptual framework. In Section III, we present the effects of the program on intertemporal choices. In Section IV, we present additional analyses that investigate the potential mechanisms underlying the treatment effects. Section V concludes.

### **I. The Financial Education Program**

The financial education program we study is offered by a German nonprofit organization, My Finance Coach (2012). Since its start-up in October 2010, it has offered financial education to over 35,000 high school students, ages 13 to 15. The program sends "finance coaches" to schools, who are employees of the (for-profit) firms that sponsor the (nonprofit) provider; they are not compensated for the training provided. They conduct several visits of 90 minutes for a total of 4.5 hours dedicated to several training modules. The provider offers a set of materials for each module and trains the coaches, so the educational program is standardized.

This program is well suited for studying the impact of financial education among adolescents for several reasons. Since it is provided at schools, all students in a class participate, avoiding selection problems (see, e.g., Meier and Sprenger 2013). Moreover, the materials are standardized, have been developed by educational experts (ranging from education researchers to school directors), and they have been extensively used for several years in Germany. Finally, this educational intervention is scalable: a large number of students can be reached using the same format of the educational program across different schools in potentially different countries.

We measure the joint impact of three training modules provided to all treated students: shopping, planning, and saving. These are the three main modules offered by the provider, who also offers other extension modules focused on topics such as the environment and business. As described in the materials supplied by the

<sup>7</sup>A study that is closely related to ours is that by Coller and Williams (1999), who provide subjects with information about market interest rates and find relatively small changes in elicited intertemporal choices. Recent studies that introduce willpower manipulations (Kuhn, Kuhn, and Villeval 2014) and savings accounts (Carvalho, Prina, and Sydnor 2016) find an increase in the frequency with which individuals choose to allocate their budgets to a single payment date. One interpretation of such results could be that such interventions also affect the extent to which experimental payments are treated as consumption and therefore the ability to identify time-preference parameters.

provider (see Table A.1 in the online Appendix for details), the shopping module focuses on the informed consumer. It emphasizes prioritizing spending (“needs and wants”), discusses criteria used in purchasing decisions, and advertising. The planning module addresses aspects of conscious planning by presenting the concepts of income and expenditure as the basis of financial planning. It also trains students on budgeting skills. The last module, saving, discusses saving motives and investment options. The training does not take a normative position on saving, but rather discusses *how* to save. Importantly, the training involves no decision directly resembling the trade-offs in the intertemporal choice task we study, which we describe in what follows.

## II. Experimental Design and Conceptual Framework

### A. Setting and Randomization

The schools in our study pertain to the two lower tracks of the German high school system. Students in these two tracks typically continue with vocational training after graduation (rather than attending college). These types of students constitute 48 percent of the high school population (Autorengruppe Bildungsberichterstattung 2016).<sup>8</sup> They have on average lower socio-economic status than students pursuing the track to attend college (Dustmann 2004).

Randomization of classes to control and treatment groups was implemented through a web interface designed by the research team. Schools in the treatment group received the training earlier in the school year, while schools in the control group received the program towards the end of the school year.<sup>9</sup>

The CTB task was conducted by the research team independently from the educational program and the coaches that had taught the materials to the students. The task was run before the schools in the control group participated in the program, and it was scheduled such that all treatment and control schools would complete the task within the same week. Randomization occurred at the school level to avoid spillover effects. Randomization was stratified by city, across the cities of Berlin, Düsseldorf, and Munich in Germany, such that differences in educational systems in the different areas are orthogonal to the treatment allocation. Since we were bound by scheduling constraints stemming from the requirement that all participating schools receive the training by the end of the school year, the time between the treatment and the intertemporal choice task was at least 4 weeks after the training and up to almost 12 weeks. We do not find differential treatment effects on students’ allocation choices in the CTB task depending on the length of the delay between the treatment and the experiment, suggesting that the effects are not short-lived.

<sup>8</sup>The school system in Germany has three types of high schools, starting as of age ten. These tracks comprise schools where students pursue vocational training (*Hauptschule*, *Sekundarschule*, *Mittelschule*), combine both vocational training with the option of attending university later on (*Realschule*, *Gesamtschule*, *Werkrealschule*), or focus on preparation for university studies (*Gymnasium*). All participating students in our study belong to the first two types of schools.

<sup>9</sup>The training took place during regular school hours. The topics of the training are not covered within the regular school curriculum.

## B. Method

The CTB task, developed by Andreoni and Sprenger (2012), asks individuals to allocate amounts of money to two points in time. The payment received at the sooner point in time,  $t$ , is  $x_t$ , while the amount received at a later point in time,  $t + k$ , is  $x_{t+k}$ . The delay between payments is  $k$ . The amounts  $x_t$  and  $x_{t+k}$  satisfy the budget constraint  $(1 + r)x_t + x_{t+k} = m$ , where  $1 + r$  is the gross interest rate. The CTB method allows for inner choices, i.e., individuals can allocate payments to both payment dates, in addition to corner solutions where the entire payment is allocated to a single payment date.

We elicit choices using three different combinations of  $t$  and  $t + k$ ; the tasks for each of these combinations are presented on a separate decision sheet. The first sheet offers payments immediately after the task finishes ( $t = 0$ , “today”) and three weeks later, so the delay is  $k = 3$  weeks. The second sheet offers payments today and six weeks later, so the delay is  $k = 6$  weeks. The last sheet offers payments in three and in six weeks, such that the delay between payments is  $k = 3$  weeks but there is also a “front-end delay” as  $t > 0$ . On each decision sheet, seven budget constraints are presented to students, where the budget  $m$  is 6 Euro. The budget we offer corresponds to about 70 percent of the average weekly income available to the students. Going from top to bottom of each decision sheet, the price for the sooner payment increases. An overview of the design appears in Table 1.<sup>10</sup>

We adapt the elicitation task, which was originally designed for college students, to students aged 13 to 15. Specifically, Andreoni and Sprenger (2012) offered a choice set with 100 choices within each budget, and in a follow-up study, Andreoni, Kuhn, and Sprenger (2015) limited the choice set to seven choices. Since they find qualitatively similar results, we simplify the presentation, reducing complexity further, to offer four combinations of sooner and later payments. In each choice situation, participants can either allocate 100 percent, 66.6 percent, 33.3 percent or 0 percent of the budget to the sooner point in time. To make the variation in the time horizons salient, color-coding is used for each point in time. Additionally, students see a calendar at the top of each sheet on which the relevant payment dates were marked in the corresponding color. An example of a decision sheet is provided in Figure 1. We randomize the ordering of the three decision sheets across classes to balance any potential order effects.

## C. Conceptual Framework

The classic model of savings and consumption (e.g., Modigliani and Brumberg 1954) assumes that fully rational consumers make optimal savings choices, smoothing marginal utility of consumption over their lifetime, given their time preferences.

<sup>10</sup>For example, for a delay of three weeks, the effective yearly interest rate, assuming quarterly compounding, ranges from 0 percent for gross rate 1.00, to 752.9 percent for gross rate 1.18, and goes up to 27,128 percent for gross rate 2.00. We chose to allow for high interest rates to capture variation in choice.





future and, in turn, decrease impatience. Based on this work, a first hypothesis is that financial education could change students' intertemporal choices by changing their time preferences. Specifically, the discussion of savings and the potential uses of money saved for the future, extensive in the planning and savings module of the financial education program, could increase a student's preference for waiting to earn the interest from saving. A direct test of this hypothesis can be conducted by eliciting incentivized intertemporal choices, using the CTB task with monetary rewards. We chose monetary rewards since they are closest to real-world financial choices, which were the object of interest in the program.

However, the identification of time preferences using monetary rewards faces three important challenges. First, it relies on the assumption that individuals display a perfect understanding of the trade-offs involved when making decisions among time-dated payments and choose the allocation that maximizes their utility given their preferences. Second, it assumes that individuals are extremely liquidity constrained and hence do not have the option to arbitrage rewards within the task with other resources outside the task. Third, if individuals have such options to arbitrage, identification of time preferences is possible only under the assumption that individuals bracket narrowly and thus do not consider these options when making choices within the CTB task.

If such assumptions are satisfied, we would expect students to display present bias and impatience and to smooth payments across time, i.e., choose interior allocations that provide positive payments on the two time periods involved. Fitting a  $\beta - \delta$  model (e.g., Laibson 1997) to such choices would imply a present-bias parameter  $\beta < 1$ , a discount factor  $\delta \leq 1$ , and utility curvature of  $\alpha < 1$ . We will show in Sections III and IV that the choices of students in the control group, who did not receive financial education, are largely consistent with such patterns, suggesting that the task was able to identify the time preferences of this group.<sup>11</sup>

If treated students display significant changes in their intertemporal allocations, this could be due to two reasons. First, the financial education program may have changed students' time preferences, as suggested by Becker and Mulligan (1997). Second, the program may have affected the validity of the assumptions required for the identification of such preferences. In what follows, we discuss these assumptions in detail.

*Choice Consistency.*—A standard assumption in models of intertemporal choice is that individuals display a perfect understanding of the trade-offs in the CTB task. Hence, they never make mistakes. Several studies have shown that individuals often exhibit violations of rationality, even in simple choice environments (see, e.g., Giné et al. 2018). One such kind of violation is the violation of the law of demand, which requires that individuals exhibit weakly more patience as the interest rate increases.<sup>12</sup>

<sup>11</sup> Several studies have found that experimental choices in time-preference elicitation tasks that use monetary rewards correlate with field behaviors related to impatience and present bias (Chabris et al. 2008; Meier and Sprenger 2010 and 2012; Castillo et al. 2011; Sutter et al. 2013).

<sup>12</sup> Precisely, within each of the three decision sheets, students made seven choices. A choice is consistent with the law of demand if the allocation to the sooner payment date decreases or stays unchanged as the interest rate



Financial education programs aim to raise awareness about the implications of consumption and savings choices in the short- and long-run. In the program we examine, adolescents discuss impulsive shopping within the shopping module; plan savings for the purchase of a durable, such as a motorcycle, in the planning module; and learn about the properties of different savings vehicles in the savings module. These discussions can improve their understanding of intertemporal trade-offs. Thus, financial education may decrease violations of the law of demand. If so, controlling for such violations will be necessary to assess the impact of the program on patience, present bias, and the share of interior choices, i.e., choices that smooth payments over time.

*Arbitrage and Bracketing.*—Students' choices only reflect their time preferences if they are extremely liquidity constrained or if they bracket their choices so narrowly in the CTB task that available borrowing and savings vehicles are ignored (e.g., Frederick, Loewenstein, and O'Donoghue 2002; Harrison, Lau, and Williams 2002; Cubitt and Read 2007; Chabris, Laibson, and Schuldt 2008; and Sprenger 2015).

Students in our sample, both in the control and treatment groups, do not appear to be extremely liquidity constrained. They receive money from their parents (about 35 Euro each month) and often have savings (about 50 percent have savings, and those who save have 71 Euro in savings, on average). Their financial situation, as measured by their income, consumption, and savings in the month preceding the experiment, does not change with the program, suggesting that background consumption was the same for both groups. Given the age of the students (13 to 15), they lack access to financial markets. However, both treatment and control students could potentially borrow from friends and family. Despite this possibility, students in the control group display present bias and choose interior allocations of the budget, consistent with significant smoothing. This suggests that the assumption of narrow bracketing is valid for students in the control group.

Financial education could change the validity of the assumption of narrow bracketing for treated students. Specifically, the degree of narrow bracketing could decrease for treated students, since the program increases students' budgeting and financial planning skills. For example, the planning module of the program raises awareness about where students' money comes from, how it is spent, and how it can be saved to achieve students' goals. By discussing their sources of income and savings, financially educated students may become more "financially sophisticated," i.e., more aware of alternative saving and borrowing opportunities.

If financial education decreases narrow bracketing, we would expect treated students to be more likely to consider the possibility of arbitraging against their alternative saving and borrowing sources. This would imply that their choices are more reflective of such sources, rather than of their time preferences. While control students exhibit present bias, treated students would be more likely to consider alternative sources for immediate consumption and hence would exhibit less present bias. At the same time, since money in the task is compared to outside

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increases. By definition, the first choice in each sheet is excluded. Thus, the fraction of choices consistent with the law of demand is the sum of consistent choices divided by 18.

sources, they would exhibit less consumption smoothing within the CTB task. Since they cannot borrow from financial markets, we would not expect their choices to reflect market interest rates. However, their choices could exhibit more patience if the transaction costs involved in borrowing from friends and family were sufficiently low.

To sum up, if students bracket narrowly throughout and the effect of the financial education treatment is to make students more patient, we would expect students to exhibit less present bias and more patience in their intertemporal choices, while equally choosing interior allocations of the budget. By contrast, if treated students are less likely to bracket narrowly due to the financial education program, we would expect such students to exhibit less present bias but also fewer interior choices. The effect of the treatment on patience would depend on the transaction costs of borrowing from informal sources, i.e., from family and friends, which are *ex ante* unknown.

To examine the effects of financial education and the interpretation of such effects, our analysis proceeds as follows. We first investigate whether financial education changes intertemporal choice. Specifically, we test the effects of financial education on time inconsistency and impatience within the CTB task. We then examine whether changes in intertemporal choices are consistent with an increase in understanding and sophistication or a change in deep time preferences.

#### D. Procedures

*Implementation of Payments.*—We follow a number of procedures to ensure trust and address issues of risk and transaction costs that typically arise when implementing delayed payments. All procedures are explained in the instructions before any decisions are taken by the students.

**Transaction Costs:** Students are given a “participation” fee of 2 Euro to thank them for their participation. They are informed that the participation fee will be split equally across both payment dates. Hence, independent of the specific choice of each student, she will always receive at least 1 Euro at each point in time.

**Record of Payments:** After students make their 21 ( $7 \times 3$ ) choices, one decision is drawn for payment to be actually implemented. The random draw is performed by one volunteer student for the entire class, and this draw is noted on the classroom board. Subsequently, based on the student’s choice and the decision drawn for payment, each student receives a payment card that records her exact payments and payment dates. Hence, students do not need to remember when the future payment would occur and how much they would receive. The payment card also serves as a written confirmation of each student’s payment entitlement. The card format is designed to fit into students’ wallets, and students are requested to keep it there. At the same time, each student writes her name on a payment list containing the payments chosen for the decision drawn for payment. This list is given to the teacher in the presence of the class. Both act as records for delayed payments, and the

payment list ensures that payments can be made even if individual payment cards are lost.

**Delivery of Payments:** Payments are made in cash to each student individually in class. Payments are made at the end of the session if today is the sooner payment date. Delayed payments are made exactly three or six weeks later in class at the dates noted on the payment cards. The exact appointment for the future payment is discussed with the teacher and then announced in class. Our instructions clearly indicate that we will return to the class once (or twice, depending on the draw) to make the delayed payments at the date(s) indicated on the calendars on the decision sheets and on payment cards. The teachers are present in class when we make this commitment.<sup>13</sup> The same procedures are followed in the control and treatment groups, so trust concerns should be the same across the groups. In line with this, we do not observe a treatment effect on the average allocation to the sooner payment date, as reported below.

**Consent:** Only students whose parents consent to participate are included in the study. The parental consent forms include the researchers' contact information, which the teacher also obtains. Our study is conducted independently from the educational program, and students, parents, and teachers are advised to contact the researchers in case of any doubts with respect to future payments. These procedures are followed in both the treatment and control groups, so as to maximize trust in both. Almost all students (97 percent) provide a signed consent form to participate in the study.

*Other Experimental Procedures.*—In each session, the instructions for the CTB task are first read aloud in front of the class.<sup>14</sup> The CTB task is conducted using pen and paper, and after the task is completed, students fill out a survey. All class visits are conducted by the same two experimenters. Students complete four basic control questions about the task before starting to provide their choices. Each student's answers are checked by the experimenters before she starts making her 21 choices.

The presentation of the instructions takes 25 minutes, on average. Students make their decisions in five to ten minutes. After they finish the CTB task, students complete a questionnaire on their gender, age, math grade, and three questions regarding their background. We elicit their household composition (i.e., who they live with), the language they speak at home, and the number of books at home. These are standard questions in the Program for International Student Assessment (PISA) survey (Frey et al. 2009) that are used to capture important family inputs into a student's education (Hanuschek and Woessmann 2011). Our survey also includes four of Raven's progressive matrices (Raven 1989), selected to measure heterogeneity in cognitive skills, based on a previous study in Germany by Heller, Kratzmeier, and Lengfelder

<sup>13</sup> Teachers are, however, kept uninformed about student choices, except for the one choice that was drawn for payment and recorded on the payment list.

<sup>14</sup> A copy of the instructions can be found in online Appendix A.

(1998). The survey also includes several questions on financial knowledge and financial behavior. The impact of the training on standard financial literacy questions is similar to the findings in Lührmann, Serra-Garcia, and Winter (2015) who study the effect of the program using survey questions in a non-experimental design. We show that financial knowledge increases among treated students in online Appendix B. We also surveyed students regarding their allowance, spending, and saving behavior. In total, the sessions last between 45 and 60 minutes.<sup>15</sup>

### E. Sample

Our sample consists of 914 students (492 in control and 422 in treatment) from 55 classes in 25 schools (12 treatment, 13 control).<sup>16</sup> The average age is 14.3 years, and 39.8 percent of the students are female. Regarding the student's family situation, we find that a substantial share, 46.4 percent, speak a language other than German at home. Also, 24 percent live with a single parent, and 60.2 percent report having fewer than 25 books at home. Individual characteristics are balanced across treatment and control, as shown by the *t*-tests presented in Table 2, supporting that randomization worked.<sup>17, 18</sup>

## III. Results: Intertemporal Choices

We first examine the impact of the financial education program on measures of impatience and present bias within the CTB task. We do not observe a significant impact of the educational program on the average allocation to the sooner payment, a measure of impatience. The average share allocated to the sooner payment date is 55.04 percent in the control group and 54.70 in the treatment group (Mann-Whitney test,  $p = 0.4795$ ). Table 3 shows the results of a regression explaining the budget share allocated to the sooner payment date. We use an interval regression model to account for the fact that students had four choices.<sup>19</sup> The coefficient of the treatment dummy is not significantly different from zero.<sup>20</sup>

<sup>15</sup> As mentioned above, in each city, all sessions were scheduled to take place during the same week for both treatment and control groups. This was possible for 46 out of 55 classes. For a small group of nine classes, the class was scheduled to be at a practical training out of school for the week, and hence, we conducted the experiment three weeks later in eight classes and six weeks later for one class. We control for any potential time effects by adding a month dummy for April (as 46 out of 55 were scheduled in April) in our regression analysis.

<sup>16</sup> While coding the answers, we found that 80 students provided multiple answers or left unanswered one or more questions. Results remain qualitatively the same if these students are included. These students are equally distributed across treatment and control ( $\chi^2$ -test,  $p$ -value = 0.43) and present no difference in observable characteristics, including gender, school grade, family situation, math grade, and cognitive ability (*t*-tests,  $p$ -value > 0.1 in all cases).

<sup>17</sup> Overall, nonresponse is very low, less than 2.4 percent of the sample. The difference in nonresponse across treatment and control is not significant for any variables, except for books at home (*t*-test,  $p$ -value = 0.04). Our results are robust to the inclusion of a dummy for nonresponse to this question.

<sup>18</sup> Throughout, we cluster standard errors at the individual level, as each individual made multiple choices in the experiment. Results remain the same if we cluster standard errors at the level of the treatment randomization, i.e., the school level.

<sup>19</sup> Results are robust to using an OLS or ordered probit regression model, as shown in online Appendix B.

<sup>20</sup> The estimated coefficient for the treatment dummy corresponds to the treatment effect for the baseline decision sheet, which elicited allocations between three and six weeks. For this baseline decision sheet we obtain a positive and nonsignificant coefficient, which can be explained by the change in the distribution of choices in the treatment group, as detailed later in this paper.

TABLE 2—INDIVIDUAL CHARACTERISTICS IN THE TREATMENT AND CONTROL GROUPS

|                       | Control | Treatment | Treatment versus control<br><i>t</i> -test ( <i>p</i> -value) |
|-----------------------|---------|-----------|---------------------------------------------------------------|
| Girl                  | 42.0%   | 37.2%     | 0.12                                                          |
| Grade 8               | 50.6%   | 52.1%     | 0.92                                                          |
| Cognition score       | 0.756   | 0.718     | 0.67                                                          |
| Math grade (relative) | 0.012   | 0.010     | 0.91                                                          |
| Migrant background    | 47.1%   | 45.7%     | 0.87                                                          |
| Single parent         | 23.4%   | 25.1%     | 0.67                                                          |
| <25 books at home     | 60.4%   | 60.1%     | 0.95                                                          |

*Notes:* This table presents the mean of the individual characteristics by treatment and control. The third column reports the *p*-value of a *t*-test that the coefficient of the treatment dummy is equal to 0 in a linear regression on each individual characteristic, using robust standard errors. Girl takes a value of 1 for female students, and grade 8 takes a value of 1 for students in that grade 8, 0 if in grade 7. Cognition score is the number of correct answers in four of Raven's progressive matrices. Math grade is defined relative to the average math grade in the class. A positive value indicates that the student performs better than the class average. Migrant background and single parent are dummy variables that take a value of 1 when the student speaks a language other than German at home and lives with a single parent, respectively. Lastly, <25 books at home is a dummy that takes a value of 1 if the subject indicated the number of books at home was either 0–10 or 11–25 (below median) and 0 if she indicated 26–100, 101–200, or more than 200 books at home (above median).

TABLE 3—DETERMINANTS OF ALLOCATION TO SOONER PAYMENT

|                                       | Allocation to sooner payment |                |
|---------------------------------------|------------------------------|----------------|
|                                       | Coefficient                  | Standard error |
| Treatment                             | 5.444                        | [4.407]        |
| Immediate payment                     | 7.726                        | [2.932]        |
| Immediate payment × treatment         | −7.057                       | [4.225]        |
| Delay is 6 weeks                      | −4.844                       | [3.031]        |
| Delay is 6 weeks × treatment          | 8.472                        | [4.358]        |
| Gross interest                        | −24.671                      | [2.284]        |
| Gross interest × treatment            | −3.147                       | [3.187]        |
| Female                                | −2.981                       | [2.014]        |
| Grade 8                               | −3.315                       | [2.232]        |
| Cognition score                       | −3.637                       | [1.215]        |
| Math grade                            | −3.720                       | [1.022]        |
| Migrant background                    | −0.873                       | [2.140]        |
| Single parent                         | 0.083                        | [2.395]        |
| <25 books at home                     | 4.641                        | [2.132]        |
| Constant                              | 78.034                       | [5.784]        |
| Observations                          | 17,724                       |                |
| Number of left-censored observations  | 4,579                        |                |
| Number of right-censored observations | 3,547                        |                |
| Number of interval observations       | 9,598                        |                |
| Pseudo log-likelihood                 | −23,721                      |                |

*Notes:* This table shows interval regression results. The dependent variable is the budget share allocated to the sooner payment date, ranging from 0 to 100. Immediate payment is a dummy variable that takes the value of 1 if the sooner payment occurred immediately after the students completed the task and survey. Delay is 6 weeks is a dummy variable that takes the value of 1 if the delay between the sooner and later payment was six weeks and not three weeks. Individual characteristics are defined as in Table 2. Month and location fixed effects are included. Interaction terms of the gross interest rate with delay as well as immediacy and their interaction with treatment are included in the regression. Robust standard errors are shown, clustered at the individual level.

The treatment group displays less present bias in its allocations than the control group. The extent of present bias within the task is measured by comparing allocation choices when the sooner payment is immediate, versus when it lies in the future. Controlling for interest rates and interaction effects, students in the control group increase their allocation by 7.73 percentage points when the sooner payment is immediate ( $p = 0.008$ ), as shown in Table 3. The effect of immediacy is reduced by 7.06 percentage points in the treatment group ( $p = 0.095$ ). A similar result is obtained by comparing the proportion of present-biased choices, as shown in Table 4. Choices are defined as present biased if students allocate a larger share of the budget to the sooner payment date when this date is immediate than when it is delayed. In the control group, on average, individuals make present-biased choices in 22.2 percent of the cases. In the treatment group, this percentage is 19.9 percent (Mann-Whitney test,  $p = 0.029$ ). The frequency of time-consistent choices, i.e., choices that are the same when the sooner payment date is immediate and when it is delayed three weeks, increases from 58.2 percent to 61.5 percent. In addition, there is a small decrease, from 19.7 percent to 18.6 percent, in the percentage of choices in which the students allocate less money when payments are immediate. In all, the distribution of present-biased, time-consistent, and future-biased choices is significantly different between treatment and control ( $\chi^2$ -test,  $p$ -value = 0.02).

The CTB task also allows the examination of sensitivity to delay, i.e., how much more impatient individuals are when the delay between the sooner and the later payment increases. In the control group, we do not find evidence of increased impatience as the delay between the sooner and later payment date increases from three to six weeks. In fact, there is a decrease of 4.84 percentage points in the allocation to the sooner payment date when the delay is six weeks instead of three.<sup>21</sup> This is somewhat surprising, as standard models of intertemporal choice assume that individuals either do not discount the future or they discount the future positively, i.e., their discount factor is weakly smaller than one. Delay sensitivity increases significantly, by 8.47 percentage points ( $p = 0.052$ ), in the treatment group, as shown in Table 3.

In Table 4, we classify decisions into three types of discounting, which allows us to assess whether it reflects a change in preferences (increased impatience) or enhanced understanding of the task. Choices are classified as consistent with positive discounting if the individual allocates a larger share of the budget to the sooner payment date when the delay is six weeks, rather than three. No discounting refers to equal allocations for both delays. A choice is classified as consistent with negative discounting if the student allocates a smaller share to the sooner payment date as delay increases. Treated students are somewhat less likely to exhibit such choices and more likely to exhibit no discounting ( $\chi^2$ -test,  $p$ -value = 0.096). Thus, the increase in delay sensitivity among treated students does not appear to be driven by a potential change in preferences, but rather increased understanding.

<sup>21</sup> Positive discounting is tested by comparing two different pairs of decision sheets. First, fixing the sooner payment date at today, we compare the allocation to the sooner payment date when the later date is in six weeks to when it is in three weeks. Second, fixing the later payment date at six weeks, we compare the allocation to the sooner payment date when the sooner payment date is today to that when it is in three weeks. Positive discounting requires a weakly positive increase in the allocation for the sooner payment date when delay is longer.



TABLE 4—TIME CONSISTENCY AND DELAY SENSITIVITY IN STUDENTS’  
INTERTEMPORAL CHOICES

|                                 | Proportion of choices |           |
|---------------------------------|-----------------------|-----------|
|                                 | Control               | Treatment |
| <i>Time consistency</i>         |                       |           |
| Present biased                  | 22.2%                 | 19.9%     |
| Time consistent                 | 58.2%                 | 61.5%     |
| Future biased                   | 19.7%                 | 18.6%     |
| $\chi^2$ -test, <i>p</i> -value | 0.020                 |           |
| <i>Delay sensitivity</i>        |                       |           |
| Positive discounting            | 19.2%                 | 19.6%     |
| No discounting                  | 61.0%                 | 62.7%     |
| Negative discounting            | 19.9%                 | 17.7%     |
| $\chi^2$ -test, <i>p</i> -value | 0.096                 |           |

*Notes:* Time-consistent choices are choices that are the same when the sooner payment date is immediate and when it is delayed three weeks. Present-biased choices are choices that allocate a larger share of the budget to the sooner payment date when this date is immediate than when it is delayed, while future-biased choices are those that allocate a smaller share to the sooner payment date when it is immediate. No discounting refers to choices that are the same when the delay is three or six weeks. Positive discounting choices are choices that allocate a larger share of the budget to the sooner payment date when the delay increases, while negative discounting choices are those that allocate a smaller share.

We further analyze understanding in the next section, by examining violations of the law of demand.

The chosen allocations vary similarly with student characteristics as found in previous studies of adolescents’ intertemporal choices. For example, we find that students with better math grades and cognition scores display more patience in their choices, in line with Castillo et al. (2011) and Sutter et al. (2013).

To sum up, we obtain two main results. First, the financial education program decreases time inconsistency in intertemporal choice. Second, it does not significantly affect the average allocation to the sooner date, a measure of patience within the CTB task. As discussed earlier, changes in intertemporal choices only reflect a change in deep preferences if individuals exhibit a complete understanding of the task and bracket narrowly, treating experimental payments as consumption. We turn to these two assumptions in the next section.<sup>22</sup>

<sup>22</sup> An alternative explanation for changes in intertemporal choices is that financial education changes adolescents’ financial situation (see, e.g., Carvalho, Meier, and Wang 2016; Dean and Sautmann 2014; and Janssens, Kramer, and Swart 2017). The survey administered after the CTB task measured the monthly allowance of each student and the amount of spending in a typical month, as well as the amount of savings in the last month. We find no significant effects of the treatment on these measures (*t*-test from a regression with a treatment dummy and robust standard errors, *p* > 0.1 for all three variables, as shown in online Appendix B). Thus, we do not find differences in students’ financial resources between the treatment and control groups that could give rise to the decrease in present bias.

TABLE 5—CONSISTENCY WITH THE LAW OF DEMAND

| <i>Dependent variable:<br/>Sample:</i> | (1)                              |                | (2)                                                |                |
|----------------------------------------|----------------------------------|----------------|----------------------------------------------------|----------------|
|                                        | Consistent choice<br>All choices |                | Allocation to sooner payment<br>Consistent choices |                |
|                                        | Coefficient                      | Standard error | Coefficient                                        | Standard error |
| Treatment                              | 0.081                            | [0.036]        | 7.309                                              | [4.946]        |
| Immediate payment                      | 0.042                            | [0.031]        | 10.115                                             | [3.492]        |
| Immediate payment $\times$ treatment   | −0.068                           | [0.047]        | −9.918                                             | [4.910]        |
| Delay is 6 weeks                       | −0.017                           | [0.031]        | −5.679                                             | [3.261]        |
| Delay is 6 weeks $\times$ treatment    | 0.041                            | [0.047]        | 9.779                                              | [4.711]        |
| Gross interest                         | 0.062                            | [0.018]        | −28.267                                            | [2.542]        |
| Gross interest $\times$ treatment      | −0.044                           | [0.028]        | −3.994                                             | [3.587]        |
| Female                                 | −0.050                           | [0.010]        | −4.419                                             | [2.334]        |
| Grade 8                                | 0.009                            | [0.012]        | −3.096                                             | [2.592]        |
| Cognition score                        | 0.008                            | [0.006]        | −4.035                                             | [1.352]        |
| Math grade                             | 0.017                            | [0.005]        | −3.738                                             | [1.195]        |
| Migrant background                     | −0.008                           | [0.011]        | −1.244                                             | [2.467]        |
| Single parent                          | −0.000                           | [0.012]        | −0.331                                             | [2.757]        |
| <25 books at home                      | −0.022                           | [0.011]        | 4.293                                              | [2.453]        |
| Constant                               |                                  |                | 78.996                                             | [6.599]        |
| Observations                           | 15,192                           |                | 14,967                                             |                |

*Notes:* Probit regression results for consistency with the law of demand are shown in column 1. Marginal effects are shown. Consistent choice takes a value of 1 if the choice is consistent with the law of demand, 0 otherwise. Interval regression results are shown in column 2, where the dependent variable is the budget share allocated to the sooner payment date, ranging from 0 to 100. The sample is restricted to those choices that are consistent with the law of demand. It also includes choices at a gross interest rate of 1, which is the lowest interest rate presented to subjects, and thus for which consistency with the law of demand cannot be tested. Individual characteristics are defined as in Table 2. Month and location fixed effects are included. Interaction terms of the gross interest rate with delay as well as immediacy and their interaction with treatment are included in all regressions. Robust standard errors are shown, clustered at the individual level.

## IV. Understanding the Treatment Effects

### A. Consistency of Choices and Present Bias

The educational program leads to an increase in consistency with the law of demand, as shown in Table 5, column 1. A choice is defined as consistent with the law of demand if share of the budget allocated to the sooner payment date weakly decreases as the interest rate increases.<sup>23</sup> The increase by 8 percentage points is significant at the 5 percent level. On average, pooling across all decisions, 80.8 percent of choices in the control and 82.9 percent in the treatment group are consistent with the law of demand. These rates are very similar to those found by Giné et al. (2018) in individual interviews with farmers in Malawi (81 percent) and

<sup>23</sup> Precisely, within each of the three decision sheets, students made seven choices. A choice is consistent with the law of demand if the allocation to the sooner payment date decreases or stays unchanged as the interest rate increases. By definition, the first choice in each sheet is excluded. Thus, the fraction of choices consistent with the law of demand is the sum of consistent choices divided by 18.

by Carvalho, Meier, and Wang (2016) in the American Life Panel, an internet sample representative of the US adult population (between 82 percent and 84 percent).<sup>24</sup>

The increase in consistency with the law of demand among treated students raises the question of whether it can explain the decrease in time inconsistency observed in choices. To answer this question, we proceed in two steps. First, we drop choices that are inconsistent with the law of demand from the data and re-estimate the treatment effects on intertemporal allocations. As shown in Table 5, column 2, we obtain qualitatively the same results. This indicates that the decrease in present bias is not driven by inconsistent choices.<sup>25</sup>

Second, the descriptive analysis suggests that, if narrow bracketing is assumed and individual preference parameters are structurally estimated, the control group exhibits time-inconsistent preferences while the treatment group does not. We examine in a structural model whether this prediction holds when we allow for errors in decision-making. If allowing for errors eliminated the change in time consistency between treatment and control, this would suggest that the change in time consistency within the CTB task is the result of errors in decision-making. Specifically, we follow Andreoni and Sprenger (2012) in assuming that individuals exhibit constant relative risk aversion (CRRA) utility within the quasi-hyperbolic discounting model (Laibson 1997). In this model, individuals discount the future with a daily discount factor,  $\delta$ , and a present-bias coefficient,  $\beta$ , that measures the degree of time inconsistency. The curvature of the period utility is determined by the parameter  $\alpha$ .<sup>26</sup>

We then extend this framework to flexibly allow for errors in decision-making. We estimate multiple error models to reduce concerns about specification error, i.e., that our results are driven by specific assumptions about the nature of stochastic choice. In the first model, we allow students to make Fechner errors, i.e., a stochastic evaluation of the distance between the optimal ratio of consumption and the available one. A larger Fechner error parameter,  $\tau$ , implies that this distance is given less weight and hence that errors are more likely (von Gaudecker, van Soest, and Wengstrom 2011).<sup>27</sup> In the second model, we follow the approach of Andersen et al. (2008), among others, and estimate a probability choice model based on Luce (1959). In this model, there is a stochastic decision error  $\sigma$ . As  $\sigma \rightarrow 0$ , choices collapse to the deterministic choice model, while as  $\sigma$  increases choices become random.

Table 6 displays the estimated parameters for each model, separately for the control and treatment groups. We first present the results for a model without errors

<sup>24</sup> In line with the idea that inconsistencies may reflect indifference between allocations, we observe higher rates of consistency with the law of demand as the offered interest rate increases.

<sup>25</sup> If we drop choices that are consistent with a negative discount rate, we obtain qualitatively similar results.

<sup>26</sup> We impose no restrictions on  $\beta$  throughout our analysis. See online Appendix C for a formal statement of the model.

<sup>27</sup> We also considered models that include a trembling-hand error in addition to Fechner or Luce errors. In these models, there is a probability  $\omega$  that an individual makes a random choice among the four available allocations of sooner and later payments (Harless and Camerer 1994). However, since such errors arise at the individual level, they would be difficult to interpret when the model is used to estimate aggregate parameters. Our preferred specification thus excludes trembling-hand errors. This being said, results remain qualitatively unchanged when we include them, as shown in online Appendix C.

TABLE 6—ESTIMATED AGGREGATE TIME-PREFERENCE PARAMETERS, BY CONTROL AND TREATMENT

| Model:<br>Group:       | Interval-censored tobit |                  | Interval-censored tobit<br>with Fechner errors |                  | Luce model       |                  |
|------------------------|-------------------------|------------------|------------------------------------------------|------------------|------------------|------------------|
|                        | Control<br>(1)          | Treatment<br>(2) | Control<br>(3)                                 | Treatment<br>(4) | Control<br>(5)   | Treatment<br>(6) |
| $\hat{\beta}$          | 0.975<br>[0.013]        | 1.003<br>[0.016] | 0.967<br>[0.017]                               | 1.005<br>[0.021] | 0.943<br>[0.018] | 0.989<br>[0.022] |
| $\hat{\delta}$         | 1.002<br>[0.001]        | 1.001<br>[0.001] | 1.003<br>[0.001]                               | 1.001<br>[0.001] | 0.991<br>[0.001] | 0.990<br>[0.001] |
| $\hat{\alpha}$         | 0.765<br>[0.021]        | 0.778<br>[0.022] | 0.802<br>[0.013]                               | 0.818<br>[0.012] | 0.821<br>[0.026] | 0.876<br>[0.027] |
| $\hat{\tau}$           |                         |                  | 2.499<br>[0.112]                               | 2.754<br>[0.140] |                  |                  |
| $\hat{\mu}$            |                         |                  |                                                |                  | 0.050<br>[0.004] | 0.059<br>[0.006] |
| Observations           | 10,332                  | 8,862            | 10,332                                         | 8,862            | 10,332           | 8,862            |
| $H_0: \hat{\beta} = 1$ | 0.0554                  | 0.8447           | 0.0537                                         | 0.8129           | 0.0021           | 0.6081           |

Notes: Columns 1 and 2 report the estimated preference parameters from the interval-censored Tobit model. Columns 3 and 4 add Fechner errors to this model. Columns 5 and 6 report the estimated preference parameters from the probability choice model, based on Luce (1959) and used in Andersen et al. (2008). All parameters are computed as nonlinear combinations, using the Delta method, of parameters estimated using maximum likelihood. Robust standard errors are presented, clustered at the individual level.

in decision making in columns 1–2. We then present the results including specific models of stochastic choice in columns 3–6. We start by discussing the parameter estimates for the control group and then turn to the treatment group in what follows.

In the control group, the estimated  $\beta$  is between 0.943 and 0.975. Throughout, it is significantly smaller than one ( $\chi^2$ -test,  $p = 0.055$  or smaller), consistent with significant present bias in students' intertemporal choices. In contrast to recent studies, we hence find present bias in the monetary domain, with slightly larger estimates than those for effort choices in Augenblick, Niederle, and Sprenger (2015). The estimated daily discount factor among control students is between 0.991 and 1.003. In two out of three specifications, students do not display significant discounting. This result reinforces our finding in the descriptive analysis that a large share (61 percent) of choices is consistent with no discounting (Table 4). The estimated CRRA parameter  $\alpha$  lies between 0.765 and 0.821 in the control group.<sup>28</sup> The value of  $\alpha$  is significantly smaller than 1 throughout, indicating that students exhibit a very high elasticity of substitution relative to previous studies (e.g., Andreoni and Sprenger 2012). These estimates imply that a \$20 split in two periods would be preferred to about \$46.5 in one period by control students (with  $\alpha = 0.821$ ). This is consistent with the high rate of interior choices in the raw data (e.g., over 48 percent of choices in the control group), which is higher than that found in other studies (e.g., 88 percent of choices are corner choices in Andreoni, Kuhn, and Sprenger 2015).

<sup>28</sup> The CRRA parameter  $\alpha$  and the Fechner error  $\tau$  are only jointly identified in the interval-censored tobit model with Fechner errors and hence should be interpreted carefully.

In the treatment group, the parameter estimate  $\hat{\beta}$  is between 0.989 and 1.005 and not significantly different from 1 ( $\chi^2$ -test,  $p > 0.1$  throughout). Hence, we do not find evidence of present bias in the choices of students in the treatment group. Relative to the control group, the estimated value of  $\beta$  increases in the treatment group ( $t$ -test,  $p = 0.058$  in the Luce model), in line with the results obtained in Table 3. This result is consistent with the interpretation that students in the control group may have perceived monetary payments more as consumption (closer to effort) than treated students.

In comparison to the control group, there is a small decrease in the discount factor among treated students. It is in line with the increased delay sensitivity found at the descriptive level, since the discount factor is identified by changes in delay sensitivity. Finally, we find that the CRRA parameter  $\alpha$  increases with program participation, though not significantly. This result is also consistent with a change in students' bracketing in the task. If control students bracket more narrowly than treated students, they should exhibit a higher elasticity of substitution. The magnitude of this change is however small, consistent with the significant but moderate decrease in the share of interior choices (8 percentage points) in the treatment group, as we discuss in the next subsection.

In summary, we find that the educational program increases the consistency of choices in the CTB task. Together with the finding that the program significantly increases consistency with zero or positive discounting, this suggests that the program increased understanding of intertemporal trade-offs. However, when we account for choice inconsistency, our results do not explain the decrease in time consistency found among treated students.

### *B. Narrow Bracketing and Present Bias*

As described earlier, a decrease in narrow bracketing would have two implications for choices within the CTB task. A first implication of a decrease in narrow bracketing would be a decrease in time inconsistency, as the one we have documented in Section III.

A second implication of a decrease in narrow bracketing is that it should reduce the degree of smoothing exhibited within the task. We refer to allocations of the entire budget to a single payment date as corner choices and examine their prevalence. In Andreoni and Sprenger (2012), around 70 percent of the choices were corner choices, suggesting that a majority of individuals may not have treated time-dated monetary payments as consumption. Corner choices are less frequent in our sample and occur more frequently in the treatment group than in the control group (48 percent versus 44 percent, respectively). Controlling for the characteristics of the budget available and individual characteristics, we find that the rate at which treated students choose corner solutions increases by close to 8 percentage points, as shown in Table 7. The fact that the rate of corner choices observed in the treatment group is almost 20 percent higher than in the control group suggests a decrease in narrow bracketing among treated students.

A further implication of a decrease in narrow bracketing is that treated students should exhibit the following pattern of choices more frequently: at low interest

TABLE 7—CORNER CHOICES

|                                      | Corner choices |                |
|--------------------------------------|----------------|----------------|
|                                      | Coefficient    | Standard error |
| Treatment                            | 0.078          | [0.043]        |
| Immediate payment                    | 0.051          | [0.029]        |
| Immediate payment $\times$ treatment | −0.025         | [0.042]        |
| Delay is 6 weeks                     | −0.032         | [0.031]        |
| Delay is 6 weeks $\times$ treatment  | 0.030          | [0.043]        |
| Gross interest                       | 0.015          | [0.019]        |
| Gross interest $\times$ treatment    | −0.026         | [0.027]        |
| Female                               | −0.175         | [0.024]        |
| Grade 8                              | 0.017          | [0.029]        |
| Cognition score                      | 0.011          | [0.015]        |
| Math grade                           | 0.005          | [0.013]        |
| Migrant background                   | −0.007         | [0.026]        |
| Single parent                        | 0.041          | [0.029]        |
| <25 books at home                    | −0.031         | [0.026]        |
| Observations                         | 17,724         |                |

*Notes:* This table shows probit regression, marginal effects, and robust standard errors clustered at the individual level. Corner choice takes a value of 1 if the choice was to allocate 0 or 100 percent of the budget to the sooner payment date. Individual characteristics are defined as in Table 2. Month and location fixed effects are included. Interaction terms of the gross interest rate with delay as well as immediacy and their interaction with treatment are included in the regression.

rates, they allocate 100 percent of the budget to the sooner payment date. As the interest rate in the CTB task increases, they switch to allocating 100 percent of the budget to the later payment date when receiving sooner payments through the CTB becomes more costly. We find a significant increase of 6 percentage points in the share of students displaying this choice pattern ( $p = 0.035$ ), as shown in online Appendix B.

Furthermore, if treated students are less likely to bracket narrowly, their choices in the task would be less informative about their deep intertemporal preferences. This would imply a weaker correlation between estimated preference parameters and intertemporal behaviors in the field, such as self-reported savings or impulsivity, among treated students. As shown in online Appendix C, we find that the correlation between estimated time-preference parameters and field savings behaviors weakly decreases with the treatment, which is consistent with the conclusion that the financial education program decreased narrow bracketing. For example, we find that in the control group, the estimated present-bias parameter ( $\hat{\beta}_i$ ) and discount factor ( $\hat{\delta}_i$ ) are correlated with the savings amount of savers. This result is consistent with the time-preference elicitation task being informative about deep intertemporal preferences among control group students. By contrast, in the treatment group, the correlation between the present-bias parameter and savings decreases significantly ( $p = 0.06$ ). It also decreases for the discount factor, though not significantly ( $p = 0.17$ ). This provides suggestive evidence that the task is less able to identify time-preference parameters among treated students.



## V. Discussion and Conclusion

This paper examines the effect of a financial education program on intertemporal choice in adolescence. Intertemporal choice is a central determinant of many important economic decisions, ranging from education, to savings, and investment behavior. Financial education may change such behaviors by increasing an individual's preference for the future or by improving her understanding of intertemporal trade-offs and sophistication in financial decision-making.

We document four main effects of financial education on intertemporal choice. Primarily, financial education increases time consistency in experimental choices among teenagers. At the same time, financial education decreases the extent of smoothing among experimental time-dated payments. It also increases consistency of choice with the law of demand, and finally, it leads to an increase in consistency with zero or positive discounting.

Taken together, these four effects suggest that financial education increases adolescents' understanding of, and sophistication in, intertemporal choice. One may ask whether there is an alternative explanation for the four effects we document. One that is common to experiments with delayed monetary payments is trust. Individuals who trust the experimenter more are typically more patient within the experimental task. We take careful precautions in the experimental design to establish trust in equal measure in both groups. Additionally, we do not observe differences in patience in our data, suggesting that the treatment effects are not due to trust.

Our findings provide new perspectives on the impact of financial education. From a methodological viewpoint, they indicate that financial education may compromise the identification of time-preference parameters using standard experimental methods. We thus contribute to the recent debate on how to measure time preferences (e.g., Sprenger 2015; and Cohen et al. 2016) by pinpointing a factor, financial education, that might be a crucial determinant of whether the restrictive assumptions needed to identify time preferences using tasks with time-dated monetary rewards are satisfied.

At the same time, our results suggest that short financial education programs may change *how* youth make intertemporal choices, enhancing their understanding and broadening the set of alternatives they consider when making such choices. Individuals who receive financial education may be more likely to behave in line with the assumptions of standard economic models, such as rationality and broad bracketing. These changes could have implications for wealth accumulation in the long run, in light of the well-established relationship between the quality of decision-making and wealth accumulation.

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